

$Z\phi$ -classes for extensions of PerfectGroup 3.A₇ = Perf(3 · 2520, 1)

Claude Archer's PhD method

Solvable quotient groups of orders $2 \leq n \leq 127$

Central: trivial coupling (central products). *Out(P)*: surjective coupling onto $\text{Out}(P) \simeq C_2$. Times are GAP runtimes. Here $|S| = 3n$ and $|E| = 7560n$.

n	$ S $	$ E $	Central	Out(P)	Total	Cumulative	Time (s)	Cumulative time (s)
2	6	15 120	1	1	2	2	0.022	0.022
3	9	22 680	2	0	2	4	0.003	0.025
4	12	30 240	2	2	4	8	0.098	0.123
5	15	37 800	1	0	1	9	0.002	0.125
6	18	45 360	3	3	6	15	0.245	0.370
7	21	52 920	1	0	1	16	0.005	0.375
8	24	60 480	5	7	12	28	0.524	0.899
9	27	68 040	6	0	6	34	0.070	0.969
10	30	75 600	2	2	4	38	0.277	1.246
11	33	83 160	1	0	1	39	0.003	1.249
12	36	90 720	8	7	15	54	1.358	2.607
13	39	98 280	1	0	1	55	0.007	2.614
14	42	105 840	2	2	4	59	0.238	2.852
15	45	113 400	2	0	2	61	0.011	2.863
16	48	120 960	14	28	42	103	3.389	6.252
17	51	128 520	1	0	1	104	0.002	6.254
18	54	136 080	11	10	21	125	1.053	7.307
19	57	143 640	1	0	1	126	0.008	7.315
20	60	151 200	5	6	11	137	0.845	8.160
21	63	158 760	4	0	4	141	0.167	8.327
22	66	166 320	2	2	4	145	0.212	8.539
23	69	173 880	1	0	1	146	0.003	8.542
24	72	181 440	22	32	54	200	3.972	12.514
25	75	189 000	2	0	2	202	0.017	12.531
26	78	196 560	2	2	4	206	0.303	12.834
27	81	204 120	22	0	22	228	0.379	13.213
28	84	211 680	4	5	9	237	0.771	13.984
29	87	219 240	1	0	1	238	0.003	13.987
30	90	226 800	6	6	12	250	0.952	14.939
31	93	234 360	1	0	1	251	0.014	14.953
32	96	241 920	51	144	195	446	22.814	37.767
33	99	249 480	2	0	2	448	0.007	37.774
34	102	257 040	2	2	4	452	0.202	37.976
35	105	264 600	1	0	1	453	0.006	37.982
36	108	272 160	33	28	61	514	4.282	42.264
37	111	279 720	1	0	1	515	0.017	42.281
38	114	287 280	2	2	4	519	0.216	42.497
39	117	294 840	4	0	4	523	0.127	42.624
40	120	302 400	14	25	39	562	4.375	46.999
41	123	309 960	1	0	1	563	0.004	47.003

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n	$ S $	$ E $	Central	Out(P)	Total	Cumulative	Time (s)	Cumulative time (s)
42	126	317 520	10	8	18	581	1.466	48.469
43	129	325 080	1	0	1	582	0.017	48.486
44	132	332 640	4	5	9	591	0.552	49.038
45	135	340 200	6	0	6	597	0.066	49.104
46	138	347 760	2	2	4	601	0.229	49.333
47	141	355 320	1	0	1	602	0.003	49.336
48	144	362 880	72	150	222	824	22.879	72.215
49	147	370 440	2	0	2	826	0.048	72.263
50	150	378 000	5	5	10	836	0.918	73.181
51	153	385 560	2	0	2	838	0.011	73.192
52	156	393 120	5	6	11	849	0.953	74.145
53	159	400 680	1	0	1	850	0.003	74.148
54	162	408 240	47	42	89	939	7.307	81.455
55	165	415 800	2	0	2	941	0.070	81.525
56	168	423 360	13	22	35	976	3.962	85.487
57	171	430 920	4	0	4	980	0.141	85.628
58	174	438 480	2	2	4	984	0.224	85.852
59	177	446 040	1	0	1	985	0.002	85.854
60	180	453 600	18	22	40	1 025	4.140	89.994
61	183	461 160	1	0	1	1 026	0.016	90.010
62	186	468 720	2	2	4	1 030	0.276	90.286
63	189	476 280	15	0	15	1 045	0.907	91.193
64	192	483 840	267	1 120	1 387	2 432	274.097	365.290
65	195	491 400	1	0	1	2 433	0.007	365.297
66	198	498 960	6	6	12	2 445	0.888	366.185
67	201	506 520	1	0	1	2 446	0.017	366.202
68	204	514 080	5	6	11	2 457	1.165	367.367
69	207	521 640	2	0	2	2 459	0.009	367.376
70	210	529 200	4	4	8	2 467	0.775	368.151
71	213	536 760	1	0	1	2 468	0.004	368.155
72	216	544 320	102	147	249	2 717	27.129	395.284
73	219	551 880	1	0	1	2 718	0.016	395.300
74	222	559 440	2	2	4	2 722	0.278	395.578
75	225	567 000	6	0	6	2 728	0.213	395.791
76	228	574 560	4	5	9	2 737	0.782	396.573
77	231	582 120	1	0	1	2 738	0.009	396.582
78	234	589 680	10	8	18	2 756	1.756	398.338
79	237	597 240	1	0	1	2 757	0.016	398.354
80	240	604 800	52	126	178	2 935	27.456	425.810
81	243	612 360	127	0	127	3 062	2.778	428.588
82	246	619 920	2	2	4	3 066	0.254	428.842
83	249	627 480	1	0	1	3 067	0.004	428.846
84	252	635 040	25	24	49	3 116	5.475	434.321
85	255	642 600	1	0	1	3 117	0.003	434.324
86	258	650 160	2	2	4	3 121	0.260	434.584
87	261	657 720	2	0	2	3 123	0.008	434.592
88	264	665 280	12	22	34	3 157	3.748	438.340

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$Z\phi$ -classes for extensions of $3.A_7$ – continued

n	$ S $	$ E $	Central	Out(P)	Total	Cumulative	Time (s)	Cumulative time (s)
89	267	672 840	1	0	1	3 158	0.003	438.343
90	270	680 400	22	20	42	3 200	4.101	442.444
91	273	687 960	1	0	1	3 201	0.050	442.494
92	276	695 520	4	5	9	3 210	0.770	443.264
93	279	703 080	4	0	4	3 214	0.218	443.482
94	282	710 640	2	2	4	3 218	0.538	444.020
95	285	718 200	1	0	1	3 219	0.011	444.031
96	288	725 760	301	958	1 259	4 478	199.508	643.539
97	291	733 320	1	0	1	4 479	0.026	643.565
98	294	740 880	5	5	10	4 489	1.446	645.011
99	297	748 440	6	0	6	4 495	0.105	645.116
100	300	756 000	16	20	36	4 531	5.484	650.600
101	303	763 560	1	0	1	4 532	0.004	650.604
102	306	771 120	6	6	12	4 544	1.109	651.713
103	309	778 680	1	0	1	4 545	0.023	651.736
104	312	786 240	14	25	39	4 584	5.673	657.409
105	315	793 800	4	0	4	4 588	0.286	657.695
106	318	801 360	2	2	4	4 592	0.304	657.999
107	321	808 920	1	0	1	4 593	0.003	658.002
108	324	816 480	153	126	279	4 872	49.694	707.696
109	327	824 040	1	0	1	4 873	0.052	707.748
110	330	831 600	6	6	12	4 885	1.760	709.508
111	333	839 160	4	0	4	4 889	0.269	709.777
112	336	846 720	43	110	153	5 042	31.883	741.660
113	339	854 280	1	0	1	5 043	0.004	741.664
114	342	861 840	10	8	18	5 061	1.900	743.564
115	345	869 400	1	0	1	5 062	0.004	743.568
116	348	876 960	5	6	11	5 073	1.084	744.652
117	351	884 520	15	0	15	5 088	1.239	745.891
118	354	892 080	2	2	4	5 092	0.308	746.199
119	357	899 640	1	0	1	5 093	0.006	746.205
120	360	907 200	61	117	178	5 271	32.571	778.776
121	363	914 760	2	0	2	5 273	0.037	778.813
122	366	922 320	2	2	4	5 277	0.343	779.156
123	369	929 880	2	0	2	5 279	0.009	779.165
124	372	937 440	4	5	9	5 288	0.842	780.007
125	375	945 000	5	0	5	5 293	0.110	780.117
126	378	952 560	44	34	78	5 371	11.882	791.999
127	381	960 120	1	0	1	5 372	0.035	792.034
Total for $2 \leq n \leq 127$			1 857	3 515	5 372	5 372	792.034	792.034